The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 January 1995

INCH-POUND

MIL-S-19500/429E
31 August 1994
SUPERSEDING
MIL-S-19500/429D
30 September 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY
TYPES 1N5615, 1N5617, 1N5619, 1N5621, 1N5623,
1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, 1N5623UL,
1N5615US, 1N5617US, 1N5619US, 1N5621US, 1N5623US,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, fast recovery power rectifier diodes. Four levels of product assurance are provided for each device as specified in MIL-S-19500. Two levels of product assurance are provided each unencapsulated device type.

1.2 Physical dimensions. See figures 1, 2, 3 (D-5A), 4, and 5 (JANHC and JANKC).

1.3 Maximum ratings.

Types	V_R	V_{RWM}	I_O		t_{rr}	T_{STG} and T_J	I_{FSM} T_A = +100°C I_O = 750 mA dc t_p = 8.3 ms	Barometric pressure (reduced)	$R_{\theta JL}$ 4/
			T_A = +55°C 1/ 2/	T_A = +100°C 2/ 3/					
	V dc	V(pk)	A dc	mA dc	ns	°C	A(pk)	mmHg	See Fig.
1N5615, 1N5615UL,US	200	200	1	750	150	-65 to +175	25	N/A	6
1N5617, 1N5617UL,US	400	400	1	750	150	-65 to +175	25	8	
1N5619, 1N5619UL,US	600	600	1	750	250	-65 to +175	25	8	
1N5621, 1N5621UL,US	800	800	1	750	300	-65 to +175	25	33	
1N5623, 1N5623UL,US	1,000	1,000	1	750	500	-65 to +175	25	33	

1/ From 1 A dc at T_A = +55°C, to .75 A dc at T_A = +100°C, derate linearly at 5.56 mA/°C.
2/ For the 1 A dc rating at +55°C and the 750 mA dc rating at +100°C, no special mounting, heat sinking, or forced-air flow across exposed areas of the device is required.
3/ From .75 A dc at T_A = +100°C, to 0 A dc at T_A = +175°C, derate linearly at 10 mA/°C.
4/ $R_{\theta JL}$ ≤ 38°C/W at L = .375 inch (9.53 mm); for UL and US suffix type devices. $R_{\theta JEC}$ = 7.0°C/W.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Research Laboratory, ATTN: AMSRL-EP-RD, Fort Monmouth, NJ, 07703-5601 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1056 Test cond. B Thermal shock (glass strain

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Method for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions shall be as specified in MIL-S-19500 and herein.

D.U.T - - - - Device under test.

JANH - - - - High reliability product assurance level for unencapsulated devices.

JANK - - - - Space reliability product assurance level for unencapsulated devices.

T_{EC} - - - - Temperature of end-caps (mounting pads for US suffix devices).

US - - - - Unleaded or surface mounted diodes with square end-caps.

UL - - - - Unleaded or surface mounted diodes with square end-caps.

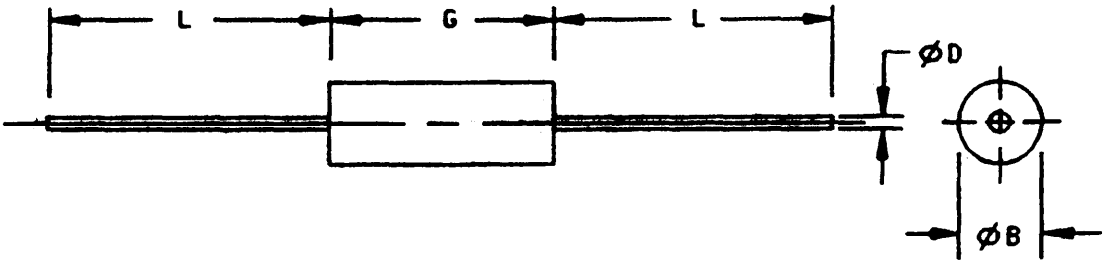
3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1, 2, 3, 4, and 5 herein. Except for JANHC and JANKC, all devices shall be metallurgically bonded-thermally-matched-noncavity-double plug construction, utilizing a category I bond, in accordance with MIL-S-19500. The UL and US version shall be structurally identical to the non-UL and non-US version except for leaded configuration.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500, and as specified herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Marking of UL and US version. For UL and US version only, all marking (see 3.5) may be omitted from the body, but shall be retained on the initial container.

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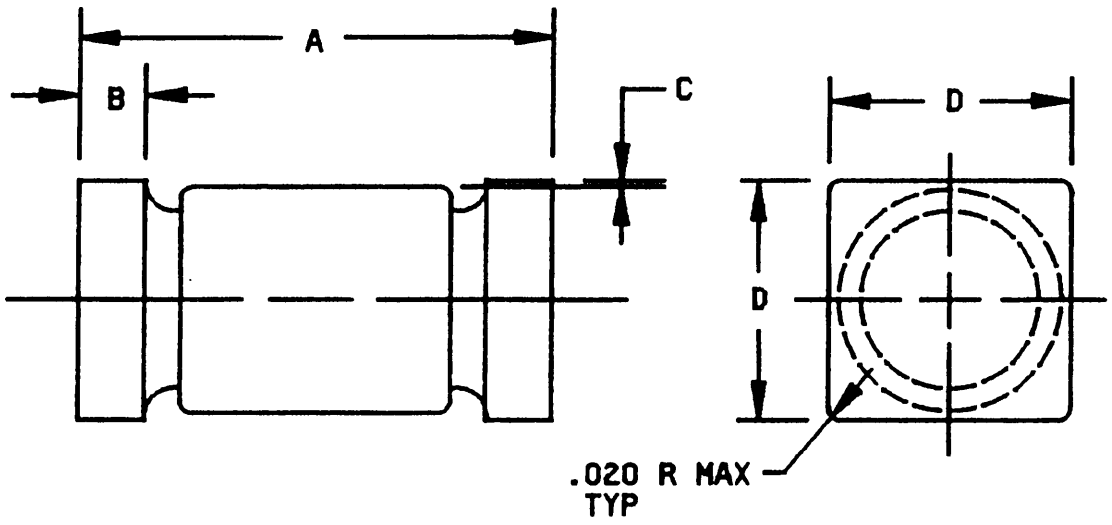


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ØB	.065	.110	1.65	2.79	3
Ød	.026	.033	.66	.84	
G	.130	.225	3.30	5.71	4
L	1.00	1.30	25.40	33.02	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension ØB shall be measured at the largest diameter.
4. Dimension G shall include all uncontrolled areas of the device.

FIGURE 1. Physical dimensions (for non-UL and non-US suffix devices only).

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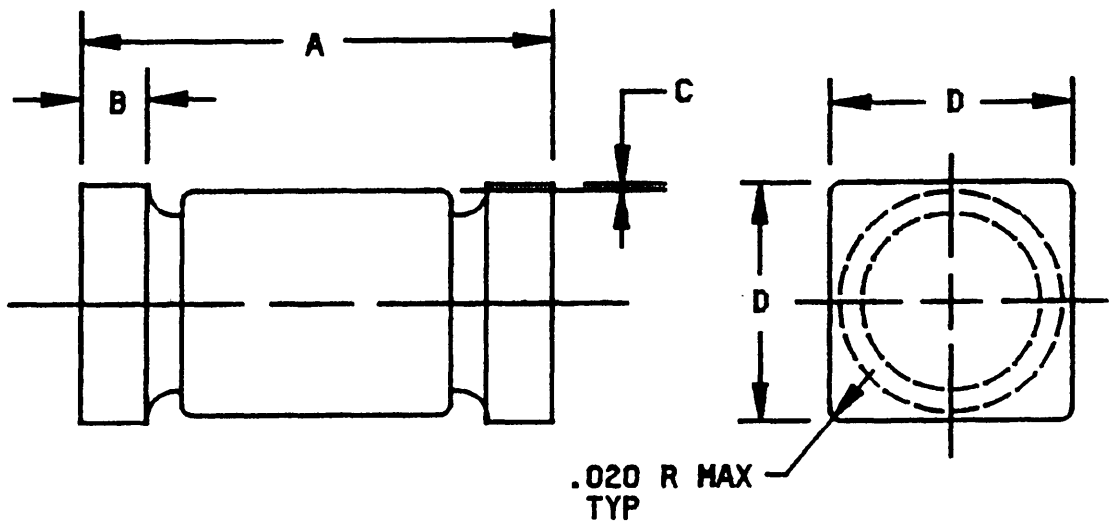


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.168	.225	4.27	5.72
B	.019	.028	0.48	0.71
C	.003	---	0.08	---
D	.091	.125	2.31	3.18

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions for types 1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, and 1N5623UL, (surface mount devices) (D-5A).

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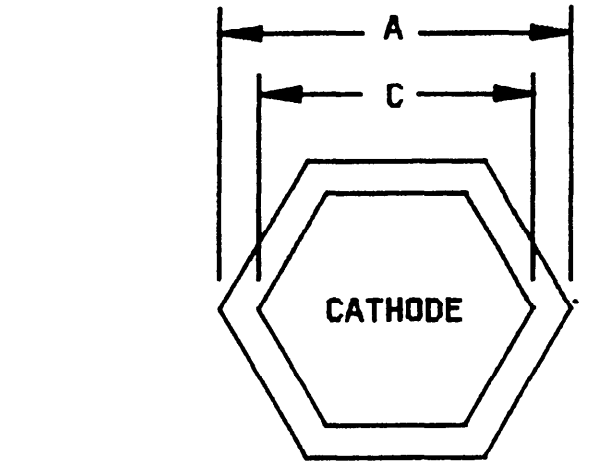


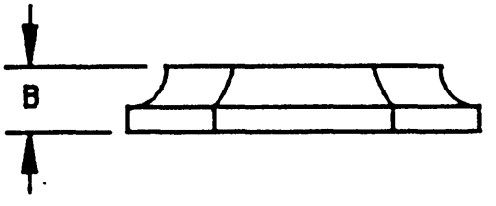
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.168	.200	4.27	5.08
B	.019	.028	0.48	0.71
C	.003	---	0.08	---
D	.091	.103	2.31	2.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Physical dimensions for types 1N5615US, 1N5617US, 1N5619US, 1N5621US, and 1N5623US, (surface mount devices) (D-5A).

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A - version

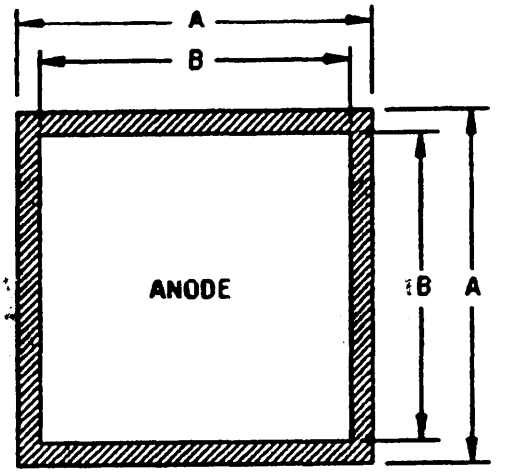
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

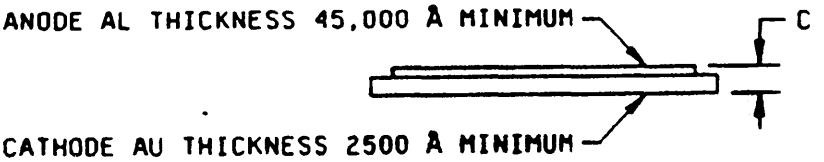
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:

Top metal: Gold 10,000 Å minimum.
Back metal: Gold 4,000 Å minimum.

FIGURE 4. Physical dimensions, JANHCA and JANKCA die.

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B - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.021	.025	0.53	0.64
C	.008	.012	0.20	0.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:

Top metal:	Aluminum (anode)	45,000 Å minimum.
Back metal:	Gold (cathode)	2,500 Å minimum.

FIGURE 5. Physical dimensions, JANHCB and JANKCB die.

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3.5 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UL and US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end or a contrasting color band may be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 JANHC and JANKC devices. Qualification for die shall be in accordance with appendix H of MIL-S-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{R1} and V_F	Not applicable
11	I_{R1} and V_F, $\Delta I_{R1} \leq$ 100 percent of initial reading or ±100 nA dc, whichever is greater. $\Delta_{VF} \leq$ ±0.1 V dc	I_{R1} and V_F
12	See 4.3.1	See 4.3.1
13 2/	Subgroups 2 and 3 of table I herein: $\Delta I_{R1} \leq$ 100 percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq$ ±0.1 V dc. Scope display evaluation (see 4.5.4).	Subgroup 2 of table I herein: $\Delta I_{R1} \leq$100 percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq$ ±0.1 V dc. Scope display evaluation (see 4.5.4).

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.

2/ $Z_{\theta JX}$ need not be repeated at step 13 if performed earlier in the screening sequence.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038, condition B of MIL-STD-750, T_A = room ambient as designated in 4.5 of MIL-STD-750, V_R = full rated V_{RWM} (see 1.3); f = 50-60 Hz; I_O = 1 A dc minimum, T_{EC} = +100°C to +125°C for surface mount devices (UL and US version).

4.3.2 Screening (JANHC or JANKC). Screening of die shall be in accordance with MIL-S-19500, appendix H.

4.3.3 Thermal impedance $Z_{\Theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101; maximum limit = 4.5°C/W.

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4.3.3.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record value $Z_{\theta JX}$) and shall be supplied to the qualifying activity on a 500 piece sample from the qualification lot prior to qualification approval.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$ in group A, subgroup 2 inspection:

a. I_M measuring current - - - - - - - 10 mA.

b. I_H forward heating current - - - - - 10 A.

c. t_H heating time - - - - - - - - - - 10 ms.

d. t_{MD} measurement delay time - - - - - 100 µs maximum.

The maximum limit for $Z_{\theta JX}$ in group A, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 and as specified in table II herein. $Z_{\theta JX}$ shall be performed after JANS group B4 only. See subgroup conditions for delta limits when applicable. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Conditions
B3	4066	I_{FSM} = 25 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, I_O = 750 mA dc; V_R = rated V_{RWM} (see 1.3); T_A = +100°C.
B4	1036	I_O = 1 A dc minimum; T_A = room ambient as defined in the general requirement of MIL-STD-750 (see 4.5); V_R = rated V_{RWM} (see 4.5.3); $t_{on} = t_{off}$ = 3 minutes minimum, 2,000 cycles; f = 50-60 Hz.
B5	1027	T_A = +150°C minimum, no heat sink, I_O = 1 A dc, minimum; adjust I_O and T_A as required to give an average T_J = +275°C +0°C, -35°C (see 4.5.3).
B6	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W, L = .375 inch (9.53 mm) (see figure 6). For surface mount devices (UL and US version) $R_{\theta JEC}$ ≤ 7°C/W.
B7	---	Peak reverse power, see 4.5.5 and figure 7 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; Sampling plan = 10, end-points, see 4.4.2.

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4.4.2.2 Group B inspection, table IVb of MIL-S-19500.

Subgroup	Method	Conditions
B2	4066	I_{FSM} = 25 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, I_O = 750 mA dc; V_R = rated V_{RWM} (see 1.3); T_A = +100°C; (T_{EC} = +100°C for surface mount devices (UL and US version)).
B3	1027	T_A = room ambient as defined in the general requirement of MIL-STD-750 (see 4.5); I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3, 4.5.3, and 4.5.6).
B5	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W, L = .375 inch (9.53 mm) (see figure 6). For surface mount device (UL and US version), $R_{\theta JEC}$ ≤ 7°C/W.
B7	---	Peak reverse power, see 4.5.5 and figure 4 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; Sampling plan = 10, end-points, see 4.4.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table II, group A, subgroup 2 and as specified in table II herein. $Z_{\theta JX}$ need not be performed. See subgroup conditions for delta limits when applicable. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
C2	2036	Tension: Test condition A, weight = 12 pounds. Lead fatigue: Test condition E, weight = 1 pound. NOTE: Both tension and lead fatigue are not applicable for surface mount device (UL and US version).
C6	1026	T_A = room ambient as defined in the general requirement of MIL-STD-750 (see 4.5); I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3, 4.5.3, and 4.5.6). T_{EC} = +100°C to +135°C for surface mount devices (UL and US version).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500.

4.4.4.1 Group E inspection, table VII of MIL-S-19500. Subgroups E1 through E4 is not applicable.

Subgroup	Method	Conditions
E5	1001	1N5617, 1N5619 = 8 mm Hg (100,000 ft); 1N5621, 1N5623 = 33 mm Hg (70,000 ft); V_R = V_{RWM} (see 1.3); t = 1 minute (minimum).

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature T_A of +25°C ±3°C.

4.5.3 Life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180°, nor less than 150°.

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4.5.4 Scope display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 5 to 20 μA/division and 50 to 200 V/division. Reverse current over the knee shall be at least 50 μA and less than 500 μA peak. Any discontinuity below actual breakdown or dynamic instability of the trace shall be cause for rejection of that device.

4.5.5 Peak reverse power test. This test shall be measured in the circuit of figure 7, or equivalent. A 20 microsecond half-sine waveform of current shall be used and peak reverse power shall be determined by the product of peak reverse voltage and peak reverse current.

4.5.6 Mounting conditions. At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that I_O is adjusted such that the junction temperature of each diode is maintained at least +120°C above the specified ambient temperature of +25°C.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 3 A dc (pulsed); (see 4.5.1) t_p = 300 µs; 2 percent maximum duty cycle	V_F	0.8	1.6	V_{pk}
Reverse current leakage	4016	DC method	I_{R1}			
1N5615, 1N5615UL,US		V_R = 200 V dc			0.5	µA dc
1N5617, 1N5617UL,US		V_R = 400 V dc			0.5	µA dc
1N5619, 1N5619UL,US		V_R = 600 V dc			0.5	µA dc
1N5621, 1N5621UL,US		V_R = 800 V dc			0.5	µA dc
1N5623, 1N5623UL,US		V_R = 1,000 V dc			0.5	µA dc
Breakdown voltage	4021		$V_{(BR)R}$			
1N5615, 1N5615UL,US		I_R = 50 µA dc		220		V dc
1N5617, 1N5617UL,US		I_R = 50 µA dc		440		V dc
1N5619, 1N5619UL,US		I_R = 50 µA dc		660		V dc
1N5621, 1N5621UL,US		I_R = 50 µA dc		880		V dc
1N5623, 1N5623UL,US		I_R = 50 µA dc		1100		V dc
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		4.5	°C/W
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC method	I_{R2}			
1N5615, 1N5615UL,US		V_R = 200 V dc			25	µA dc
1N5617, 1N5617UL,US		V_R = 400 V dc			25	µA dc
1N5619, 1N5619UL,US		V_R = 600 V dc			25	µA dc
1N5621, 1N5621UL,US		V_R = 800 V dc			25	µA dc
1N5623, 1N5623UL,US		V_R = 1,000 V dc			25	µA dc

See footnote at end of table.

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TABLE 1. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Reverse recovery time	4031	Condition B1	t_{rr}			
1N5615, 1N5615UL,US					150	ns
1N5617, 1N5617UL,US					150	ns
1N5619, 1N5619UL,US					250	ns
1N5621, 1N5621UL,US					300	ns
1N5623, 1N5623UL,US					500	ns
Capacitance	4001	V_R = 12 V dc 0.1 < f < 1.0 MHz	C			
1N5615, 1N5615UL,US					45	pF
1N5617, 1N5617UL,US					35	pF
1N5619, 1N5619UL,US					25	pF
1N5621, 1N5621UL,US					20	pF
1N5623, 1N5623UL,US					15	pF
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Reverse leakage current change	4016	DC method	ΔI_{R1} 4/		For JANTX and JANTXV, ≤250 nA dc or 100 percent, whichever is greater; for JANS, ≤ 100 nA dc or 100 percent, whichever is greater.	
2	Forward voltage change	4011	I_F = 3 A dc; Pulsed (see 4.5.1) t_p = 300 µs; 2 percent maximum duty cycle.	ΔV_{F1} 4/		± 50 mV dc maximum change from previous measured value.	

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, step 2.

b. Subgroup 4, see table II herein, step 2.

c. Subgroup 5, see table II herein, step 1.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, step 1.

b. Subgroup 6, see table II herein, step 1.

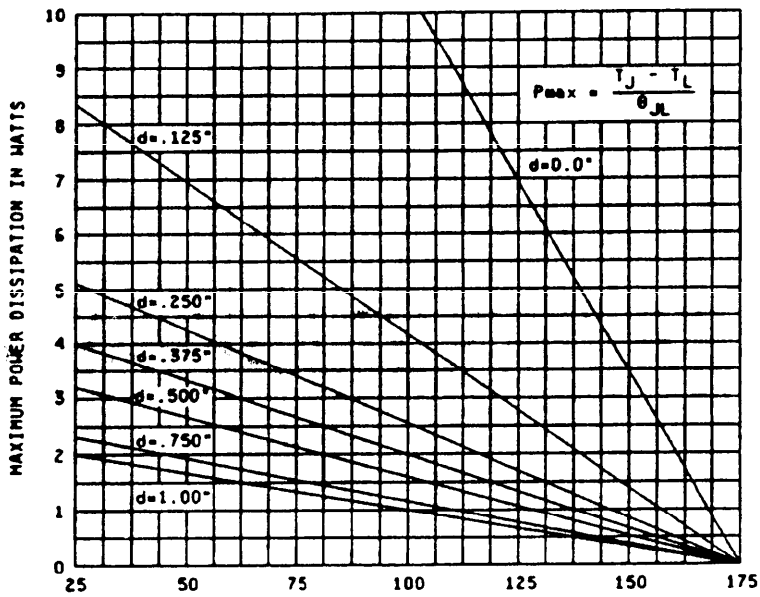
3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, step 1 (JANS).

b. Subgroup 6, see table II herein, step 1 and 2 (JANS), step 1 (JANTX and JANTXV).

4/ Devices which exceed the group A limits for this test shall not be accepted.

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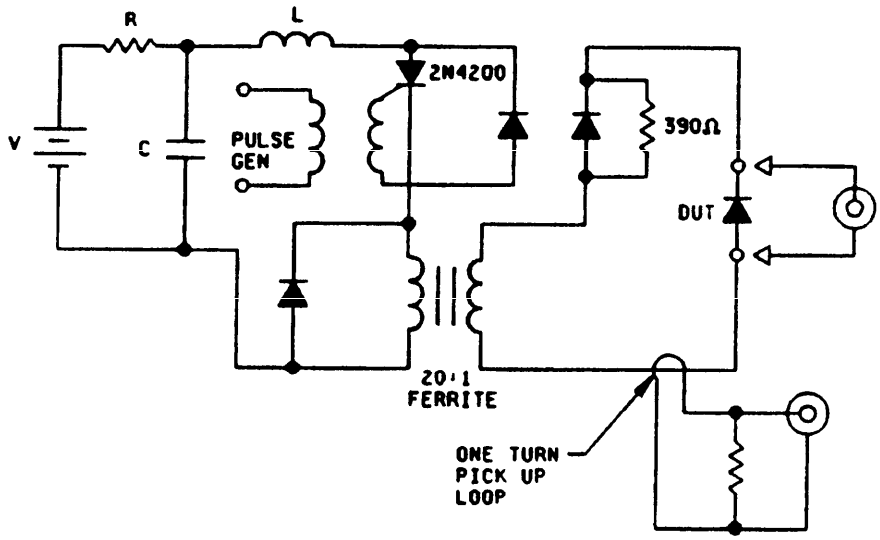
Maximum lead temperature in °C (T_L) at point d from body (see table below).
For maximum operating junction temperature of +175°C with equal two-lead conditions.

d (see notes 1 and 2)		$R_{\theta JL}$
Dimensions		°C/W
Inches	mm	
.000	(0.00)	7
.125	(3.18)	18
.250	(6.35)	29
.375	(9.53)	38
.500	(12.70)	45
.750	(19.05)	62
1.000	(25.40)	76

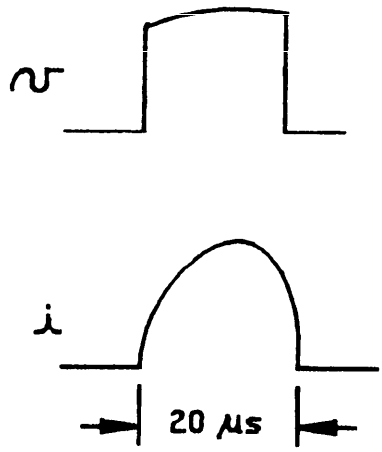
NOTES:
1. Lead distance point d and thermal resistance data.
2. Thermal resistance test performed at .375 inch (9.53 mm).
3. $R_{\theta JEC} \leq 7°C/W$.

FIGURE 6. Maximum power versus lead temperature.

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NOTES:
1. L - 13T #22 pm 1 inch (25.4 mm) diameter form (air core).
2. C - 1 to 10 μfd to give a 20 μs pulse width.
3. V - adjustable to 200 volts for power desired in D.U.T.



TYPICAL WAVE FORMS

FIGURE 7. Peak reverse power measurement circuit and waveform.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level, type designator, and for die acquisition, the JANHC and JANKC identification (see 6.3 and figures 4 and 5).

6.3 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5615) will be identified on the Qualified Products List (QPL).

JANHC and JANKC ordering information		
PIN	Manufacturer	
	14552	33178
1N5615	JANHCA1N5615	JANHCB1N5615
	JANKCA1N5615	JANKCB1N5615
1N5617	JANHCA1N5617	JANHCB1N5615
	JANKCA1N5617	JANKCB1N5615
1N5619	JANHCA1N5619	JANHCB1N5615
	JANKCA1N5619	JANKCB1N5615
1N5621	JANHCA1N5621	JANHCB1N5615
	JANKCA1N5621	JANKCB1N5615
1N5623	JANHCA1N5623	JANHCB1N5615
	JANKCA1N5623	JANKCB1N5615

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1610)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/429E	2. DOCUMENT DATE (YYMMDD) 940831

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY TYPES 1N5615, 1N5617, 1N5619, 1N5621, 1N5623, 1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, 1N5623UL, 1N5615US, 1N5617US, 1N5619US, 1N5621US, 1N5623US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. TYPE OF ORGANIZATION VENDOR MANUFACTURER USER OTHER (Specify)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
DIRECTOR, US ARMY RESEARCH LAB/EPSD,	(1) Commercial 908-544-3441/3148	(2) AUTOVON 995-3441/3148
c. ADDRESS (Include Zip Code) ATTN: AMSRL-EP-RD FORT MONMOUTH, NJ 07703-5601	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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